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Department of Computer Science and Engineering

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DATA STRUCTURES AND ALGORITHMS-3
(25CS2103E)

CitationFlow Analytics

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CITATIONFLOW ANALYTICS

1. Case Study / Introduction

CitationFlow Analytics is a Data Structures and Algorithms project that models research papers as a directed citation graph. Each research paper is represented as a vertex, while a citation is represented as a directed edge from the citing paper to the referenced paper. The graph model allows the application to study connectivity, citation paths, citation-based ranking and network flow.

The project uses a small curated dataset of real publications from computer vision and deep learning. The application does not automatically download papers from Google Scholar at runtime. Instead, the selected paper metadata and citation relationships are included as a demonstration dataset so that the DSA algorithms can be executed reliably during a presentation.

2. Problem Statement

Develop a citation analysis system that models research papers as a directed citation network and analyzes citation relationships using graph and network-flow algorithms. The system should identify citation-based influential papers, find citation paths, calculate maximum modeled citation flow between selected papers, visualize the network and generate an analytical report.

Input: Directed citation graph with positive edge capacities.

Output: Maximum modeled citation flow, ranked papers, citation paths, traversal results, visualization and report.

Constraint: Directed graph; positive capacities.

3. Objectives

- Represent research papers and citation relationships using a directed graph.
- Store the graph using an adjacency-list representation.
- Apply BFS and DFS for graph traversal.
- Find directed citation paths between selected papers.
- Rank papers using incoming citation count (in-degree) and a PriorityQueue.
- Calculate maximum modeled citation flow using Edmonds-Karp.
- Allow papers and citation relationships to be added during a running session.
- Visualize the citation network and generate an analytical report.

4. Dataset

The demonstration uses real publications from computer vision and deep learning literature. Their metadata is embedded in the project; the network itself is a small curated classroom dataset.

ID	Research Paper	Year
P1	ImageNet: A Large-Scale Hierarchical Image Database	2009
P2	ImageNet Classification with Deep Convolutional Neural Networks	2012
P3	Very Deep Convolutional Networks for Large-Scale Image Recognition	2014
P4	Deep Residual Learning for Image Recognition	2016
P5	Attention Is All You Need	2017
P6	An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale	2021
P7	Swin Transformer: Hierarchical Vision Transformer using Shifted Windows	2021
P8	Learning Transferable Visual Models From	2021

5. System Architecture

Layer	Technology / Component	Purpose
Frontend	HTML, CSS, JavaScript	Dashboard and user interaction
Backend	Java HttpServer	API endpoints and DSA execution
Graph	HashMap / LinkedHashMap + Lists	Adjacency-list representation
Traversal	BFS / DFS	Graph traversal and path search
Ranking	In-degree + PriorityQueue	Citation-based ranking
Flow	Edmonds-Karp	Maximum modeled flow
Storage	In-memory Java collections	Current-session data

6. Citation, Capacity and Maximum Citation Flow

Citation: a directed relationship Citing Paper $\rightarrow$ Referenced Paper. For example, $P7 \rightarrow P4$ means $P7$ cites $P4$ in the selected network.

Capacity: a positive project-defined flow value attached to a citation edge. The seeded demonstration uses capacity 1 for citation edges. This is NOT the Google Scholar citation count; it is the capacity used by the network-flow model.

Maximum modeled citation flow: the largest amount of flow that can be sent from a selected source paper to a selected target paper without exceeding the capacities of the directed edges. In the seeded demonstration, source $P7$ and target $P1$ produce a maximum modeled flow of 3.

7. Frontend Features

- Research Paper Dataset
- Directed Citation Graph
- BFS and DFS traversal
- Citation Path search
- Maximum Citation Flow
- Influential Papers / ranking
- Add Research Paper
- Create Citation
- Analytical Report

8. Creating a Citation

If a new paper $P9$ is added and the user creates $P9 \rightarrow P7$ with capacity 1, the system interprets the relationship as $P9$ cites $P7$. The new edge is added to the adjacency list and can be used by later graph operations.

9. Expected Outcome

Analyze citation networks, identify influential papers, calculate citation flow, and visualize research relationships.

10. Technology Choice

Java and the Java Collections Framework are used for the DSA implementation. HTML/CSS/JavaScript provide the frontend.

ALGORITHM / PSEUDO CODE

1. Directed Graph and Adjacency List

Graph: Map<String, List<Edge>>
Edge = {from, to, capacity}

ADD-CITATION(u, v, c):

1. Check that u and v exist.
2. Check $u \neq v$.
3. Check $c > 0$.
4. Check that $u \rightarrow v$ is not already present.
5. Add Edge(u, v, c) to graph[u].

2. BFS

BFS(start):

1. Create queue Q and visited set.
2. Add start to Q and visited.
3. While Q is not empty:
 - a. $u = Q.remove()$.
 - b. Record u.
 - c. For each edge $u \rightarrow v$: if v is unvisited, visit and enqueue v.
4. Return traversal order.

3. DFS

DFS(u):

1. Mark u visited.
2. Record u.
3. For every edge $u \rightarrow v$: if v is unvisited, call DFS(v).
4. Return after all neighbors are processed.

4. Citation Path using BFS

FIND-PATH(source, target):

1. Run BFS from source.
2. When v is first discovered from u, set $parent[v] = u$.
3. Stop when target is found or no vertices remain.
4. If target is not found, report no path.
5. Follow parent[] from target to source.
6. Reverse the sequence and display the directed path.

5. In-Degree Ranking

In-degree is the number of incoming citation edges. It is used as a simple citation-based influence measure within the selected network.

RANK-PAPERS:

1. Set $\text{indegree}[p] = 0$ for every paper p .
2. For each directed edge $u \rightarrow v$, increment $\text{indegree}[v]$.
3. Insert papers into a PriorityQueue ordered by descending in-degree.
4. Remove items from the queue to obtain ranking order.
5. Display ID, title and in-degree.

6. Edmonds-Karp Maximum Flow

MAX-FLOW(source, target):

1. Build residual capacities from the citation edges.
2. Set $\text{totalFlow} = 0$.
3. Repeat:
 - a. Use BFS to find source $\rightarrow$ target with residual capacity > 0 .
 - b. If no augmenting path exists, stop.
 - c. $\text{pathFlow} = \text{minimum residual capacity on that path}$.
 - d. $\text{totalFlow} += \text{pathFlow}$.
 - e. Update forward and reverse residual capacities.
4. Return totalFlow .

7. Why BFS is Used

BFS is used for normal traversal because it visits vertices level by level. It is also used inside Edmonds-Karp to find augmenting paths in the residual graph.

8. Example

For source P7 and target P1, the seeded network contains multiple directed routes. Edmonds-Karp repeatedly finds usable augmenting paths and updates residual capacities. The resulting maximum modeled flow is 3. This is a flow value based on project-defined capacities, not a claim about the actual number of citations between the two papers.

9. Complexity

Operation	Complexity / Note
BFS	$O(V + E)$
DFS	$O(V + E)$
In-degree calculation	$O(V + E)$
PriorityQueue ranking	$O(V \log V)$ after in-degree
Edmonds-Karp	$O(VEZ)$ standard worst-case bound
Adjacency-list storage	$O(V + E)$ space

CODE

The following excerpts are taken directly from the submitted project source file src/Main.java. They show the actual graph representation, BFS traversal, Edmonds-Karp maximum-flow implementation, PriorityQueue ranking and dynamic citation creation.

1. Edge and Graph Representation

```
static class Edge {
    String from, to; int capacity;
    Edge(String from, String to, int capacity) {
        this.from=from; this.to=to; this.capacity=capacity;
    }
}
static Map<String,Paper> papers = new LinkedHashMap<>();
static Map<String,List<Edge>> graph = new LinkedHashMap<>();
```

2. BFS Traversal

```
static List<String> bfs(String start){
    List<String> order=new ArrayList<>(); if(!papers.containsKey(start))return order;
    Queue<String> q=new ArrayDeque<>(); Set<String> seen=new HashSet<>();
    q.add(start); seen.add(start);
    while(!q.isEmpty()){
        String u=q.poll(); order.add(u);
        for(String v:neighbors(u)) if(seen.add(v)) q.add(v);
    }
    return order;
}
```

3. PriorityQueue Ranking

```
PriorityQueue<String> pq=new PriorityQueue<>((a,b)->{
    int c=Integer.compare(indegree.get(b),indegree.get(a));
    return c!=0?c:a.compareTo(b);
});
pq.addAll(papers.keySet());
while(!pq.isEmpty()) out.add(pq.poll());
```

4. Edmonds-Karp Maximum Flow

```
static FlowResult maxFlow(String source,String sink){
    FlowResult out=new FlowResult();
    if(!papers.containsKey(source) || !papers.containsKey(sink) || source.equals(sink))return out;
    List<String> ids=new ArrayList<>(papers.keySet());
    Map<String,Integer> index=new HashMap<>();
    for(int i=0;i<ids.size();i++) index.put(ids.get(i),i);
    int n=ids.size(); int[][] residual=new int[n][n];
    for(List<Edge> list:graph.values())
        for(Edge e:list) residual[index.get(e.from)][index.get(e.to)]+=e.capacity;
    int s=index.get(source), t=index.get(sink);
    while(true){
        int[] parent=new int[n]; Arrays.fill(parent,-1); parent[s]=s;
        Queue<Integer> q=new ArrayDeque<>(); q.add(s);
        while(!q.isEmpty()&&parent[t]==-1){
            int u=q.poll();
            for(int v=0;v<n;v++){
                if(parent[v]==-1&&residual[u][v]>0){
                    parent[v]=u; q.add(v);
                    if(v==t) break;
                }
            }
        }
        if(parent[t]==-1) break;
        int bottleneck=Integer.MAX_VALUE;
```

```

        for(int v=t;v!=s;v=parent[v])
            bottleneck=Math.min(bottleneck,residual[parent[v]][v]);
        out.maxFlow+=bottleneck;
        for(int v=t;v!=s;v=parent[v]){
            int u=parent[v];
            residual[u][v]-=bottleneck;
            residual[v][u]+=bottleneck;
        }
    }
    return out;
}

```

5. Dynamic Citation Creation

```

static boolean addEdge(String from,String to,int capacity) {
    if(!papers.containsKey(from)||!papers.containsKey(to)||
        from.equals(to)||capacity<=0||hasEdge(from,to)) return false;
    graph.get(from).add(new Edge(from,to,capacity));
    return true;
}

```

6. Frontend / Backend Integration

The Java `HttpServer` exposes endpoints for papers, graph, BFS, DFS, citation path, maximum flow, ranking, report generation, adding papers and adding citations. The JavaScript frontend in `web/index.html` calls these endpoints and displays the results.

RESULTS



CitationFlow Analytics

Research Citation Network Analysis & Influence Ranking System

DSA • Directed Graph • Network Flow

8

Research papers (V)

15

Citation edges (E)

P1

Top paper by in-degree

3

Demo maximum flow

$O(VE^2)$

Edmonds-Karp

01 • MODEL

Directed Citation Graph

A research paper is a vertex. A citation is a directed edge from the **citing** paper to the **referenced** paper. Every edge has a positive capacity for the network-flow model.

P2 → P1

ImageNet Classification with Deep Convolutional Neural Networks → ImageNet: A large-scale hierarchical image database
capacity = 1

P4 → P2

Deep Residual Learning for Image Recognition → ImageNet Classification with Deep Convolutional Neural Networks
capacity = 1

P4 → P3

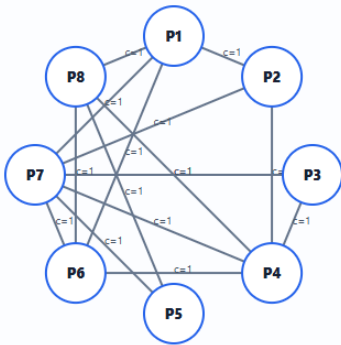
Deep Residual Learning for Image Recognition → Very Deep Convolutional Networks for Large-Scale Image Recognition
capacity = 1

P6 → P4

02 • VISUALIZATION

Citation Network Map

The graph view makes the relationship structure easy to explain during the viva.



03 • DATA

Research Paper Dataset

The seeded records are real publications. The project uses a selected citation network for demonstration; it is not intended to reproduce the entire academic citation database.

ID	Research paper	Authors	Year	Manage
P1	ImageNet: A large-scale hierarchical image database	J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei	2009	Delete
P2	ImageNet Classification with Deep Convolutional Neural Networks	A. Krizhevsky, I. Sutskever, G. E. Hinton	2012	Delete

03 • DATA

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ID	Research paper	Authors	Year	Manage
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P2	ImageNet Classification with Deep Convolutional Neural Networks	A. Krizhevsky, I. Sutskever, G. E. Hinton	2012	Delete
P3	Very Deep Convolutional Networks for Large-Scale Image Recognition	K. Simonyan, A. Zisserman	2014	Delete
P4	Deep Residual Learning for Image Recognition	K. He, X. Zhang, S. Ren, J. Sun	2016	Delete
P5	Attention Is All You Need	A. Vaswani, N. Shazeer, N. Parmar et al.	2017	Delete
P6	An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale	A. Dosovitskiy, L. Beyer, A. Kolesnikov et al.	2021	Delete
P7	Swin Transformer: Hierarchical Vision Transformer using Shifted Windows	Z. Liu, Y. Lin, Y. Cao, H. Hu et al.	2021	Delete
P8	Learning Transferable Visual Models From Natural Language Supervision	A. Radford, J. W. Kim, C. Hallacy et al.	2021	Delete

04 • TRAVERSAL

BFS & DFS

BFS explores level-by-level using a queue. DFS explores deeply before backtracking. Both traverse the adjacency list in $O(V + E)$.

Start paper

P6 — An Image is Worth 16x16 Words: Transformers for

Run BFS

Run DFS

BFS order: P6 → P4 → P1 → P2 → P3

05 • PATH ANALYSIS

Citation Path

Find a directed path from one paper to another using BFS and parent tracking.

Source / citing paper

P7 — Swin Transformer: Hierarchical Vision Transformer

Target / referenced paper

P1 — ImageNet: A large-scale hierarchical image databa

Find Path

Path: P7 → P1

Edges traversed: 1

06 • NETWORK FLOW

Maximum Citation Flow

Uses **Edmonds-Karp**, a Ford-Fulkerson implementation that uses BFS to find augmenting paths in the residual graph.

Source

P7 — Swin Transformer: Hierarchical Vision Transformer

Target

P1 — ImageNet: A large-scale hierarchical image databa

Calculate Maximum Flow

3

Maximum modeled flow: P7 → P1

1. P7 → P1 bottleneck 1

2. P7 → P2 → P1 bottleneck 1

3. P7 → P6 → P1 bottleneck 1

Capacity is a project-defined flow unit, not a Google Scholar citation count.

07 • RANKING

Influential Papers

The project ranks papers by **in-degree**, i.e. the number of incoming citation edges inside this selected network. A Java **PriorityQueue** produces the ranking.

07 • RANKING

Influential Papers

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Rank	ID	Paper	Incoming citations
#1	P1	ImageNet: A large-scale hierarchical image database	4
#2	P4	Deep Residual Learning for Image Recognition	3
#3	P2	ImageNet Classification with Deep Convolutional Neural Networks	2
#4	P3	Very Deep Convolutional Networks for Large-Scale Image Recognition	2
#5	P5	Attention Is All You Need	2
#6	P6	An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale	2
#7	P7	Swin Transformer: Hierarchical Vision Transformer using Shifted Windows	0
#8	P8	Learning Transferable Visual Models From Natural Language Supervision	0

08 • MANAGEMENT

Add Research Paper

Adds a new graph vertex to the current Java session.

Demo: use P9, then create P9 → P7 in the citation form.

ID

P9

Title

Research paper title

Year

2026

Add Paper

09 • RELATIONSHIP

Create Citation

From = citing paper → To = referenced paper. Capacity must be a positive integer.

From

P7 — Swin Transformer: Hierarchical Vision Transformer using Shifted Windows

To

P1 — ImageNet: A large-scale hierarchical image database

Capacity

10 • COMPLEXITY

Algorithm & Scalability Analysis

These complexity values directly support the computational-requirements discussion in the DSA PPT.

BFS O(V + E) Queue + visited set	DFS O(V + E) Recursion + visited set	Path BFS O(V + E) Parent reconstruction	Ranking O(E + V log V) Indegree + PriorityQueue	Edmonds-Karp O(VE ²) Residual BFS iterations
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11 • REPORT

Generated Citation Analysis Report

Current network: 8 papers, 15 directed citation edges.

Top in-degree paper: P1 — ImageNet: A large-scale hierarchical image database (4 incoming edges).

Demo flow: P7 → P1 = 3 modeled flow units.

Algorithms: Adjacency List • BFS • DFS • BFS Path • PriorityQueue Ranking • Edmonds-Karp Maximum Flow.

Complexities: BFS/DFS/Path O(V+E); Ranking O(E+V log V); Edmonds-Karp O(VE²).

Interpretation: this system demonstrates algorithmic citation-network analysis on a selected real-paper dataset. Edge capacity and in-degree are project metrics, not authoritative academic impact metrics.

Download Report (.txt)

